

Structural Reconstruction of the Voynich Manuscript

Conclusion

The synthesized structural field has three layers. The **base transition field** is the residual left by the phi-transition predictor. The **bifolio-band component** is the inverse-Fourier reconstruction of the two dominant boundary-enhanced bands at scale rungs 9.36–9.72. The **secondary harmonic** is the retained higher-scale component at rung 15.48. Their sum reconstructs the measured residual field to floating-point precision.

The result is a structural reconstruction of symbol transitions and codicological periodicities. It is not a plaintext translation and does not assign meanings to Voynich symbols.

Model

The transition generator is:

$$Tx[n] = \phi^{-1}x[n - 1] + \phi^{-2}x[n + 1].$$

The residual energy field is:

$$\varepsilon[n] = \|x[n] - Tx[n]\|^2.$$

The synthesized field is:

$$\varepsilon_{syn}[n] = \varepsilon_{base}[n] + \varepsilon_{bifolio}[n] + \varepsilon_{harmonic}[n].$$

Measured components

Component	Scale	Variance or energy	Structural role
Base transition residual	broadband	residual variance 0.1295487122	local symbol departures from the phi-transition generator
Bifolio band	scale rungs 9.36–9.72	variance 0.0038701360;	repeated gathering-scale component

		2.9874% of residual energy	
Secondary harmonic	scale rung 15.48	variance 0.0001402350; 0.1082% of residual energy	higher-scale harmonic

The primary filtered component has boundary correlation 0.0034309637. Its mean is 0.0006361028 within 100 symbols of modeled bifolio starts and -0.0000369276 in the interior.

Codicological alignment

The documented manuscript model treats a standard quire as four bifolios or eight folios. The modeled bifolio boundary test used 52 boundaries. The near-boundary residual mean was 1.4330394734 compared with 1.4239211097 in the interior, with permutation $p=0.0075$. The quire model produced $p=0.0345$. The bifolio-scale result is therefore the stronger of the two modeled gathering signals.

Reconstructed structure

The combined field supports the following interpretation. A long-range phi-weighted transition field organizes the symbol stream. A smaller but reproducible band-limited component is enhanced near modeled bifolio transitions. A weaker higher harmonic persists at a larger scale. The manuscript's folio order and codicological transitions therefore appear in separate spectral layers rather than as one single period.

Boundary of the reconstruction

The RF transcription provides folio and locus order, not image-plane coordinates. The reconstruction therefore maps the residual field to codicological sequence, not to manuscript pixels. Exact physical gathering assignments, foldout topology, and image-coordinate registration require a complete collation table and page-coordinate dataset.

Artifacts

The synthesized arrays are stored in `voynich_structural_reconstruction.npz` with keys `base_transition` , `bifolio_band` , `secondary_harmonic` , and `synthesized_residual` .

References

- [1] Description of the Voynich Manuscript
- [2] Voynich Manuscript Transcription Resources